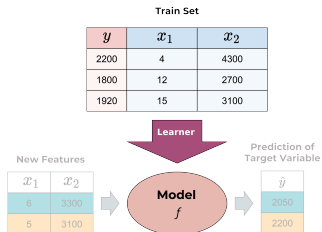
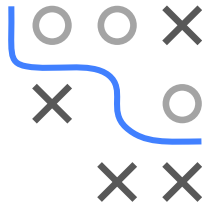


Introduction to Machine Learning

ML-Basics Learner

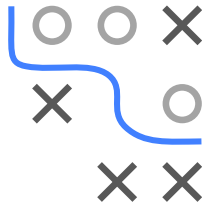


Learning goals

- Know formal definition of learner
- Understand that a learner receives training data and outputs the best model from $\mathcal{H}$

SUPERVISED LEARNING EXAMPLE

- Imagine we want to investigate how working conditions affect productivity of employees
- It is a **regression** task since the target *productivity* is continuous
- We collect data about worked minutes per week (*productivity*), how many people work in the same office as the employee in question, and the employee's salary



Features x		Target y
People in Office (Feature 1) x_1	Salary (Feature 2) x_2	Worked Minutes Week (Target Variable)
4	4300 €	2220
12	2700 €	1800
5	3100 €	1920

Diagram illustrating a dataset with 3 samples ($n = 3$) and 2 features ($p = 2$).

The features are:

- People in Office (Feature 1) x_1
- Salary (Feature 2) x_2

The target variable is:

- Worked Minutes Week (Target Variable) y

The data points are:

- Sample 1: $x_1 = 4$, $x_2 = 4300$ €, $y = 2220$
- Sample 2: $x_1 = 12$, $x_2 = 2700$ €, $y = 1800$
- Sample 3: $x_1 = 5$, $x_2 = 3100$ €, $y = 1920$

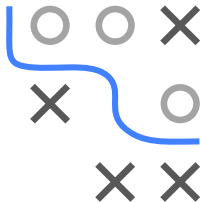
SUPERVISED LEARNING EXAMPLE

How could we construct a model from these data?

We could investigate the data manually and come up with a simple, hand-crafted rule such as:

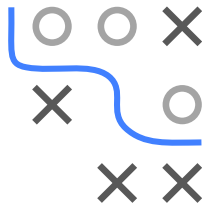
- The baseline productivity of an employee with salary 3000 and 7 people in the office is 1850 minutes
- A decrease of 1 person in the office increases productivity by 30
- An increase of the salary by 100 increases productivity by 10

⇒ Obviously, this is neither feasible nor leads to a good model.



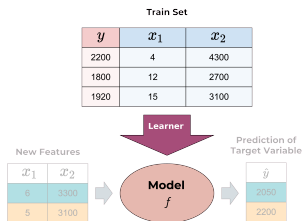
IDEA OF SUPERVISED LEARNING

- **Goal:** Identify the functional relationship that maps features to target
- **Supervised** learning means we use *labeled* data to learn model f
- Later, we use model f to predict y for new *unlabeled* data



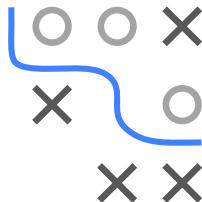
LEARNER DEFINITION

- Algorithm for finding f is called **learner** / **learning algorithm** / **inducer**
- The learner is our means of picking the best element from the hypothesis space $\mathcal{H}$ for given training data
- Formally it maps training data $\mathcal{D} \in \mathbb{D}$ (plus a vector of **hyperparameter** control settings $\lambda \in \Lambda$) to a model:



$$\mathcal{I} : \mathbb{D} \times \Lambda \rightarrow \mathcal{H}$$

- Practically, we often construct a mapping $\mathcal{I} : \mathbb{D} \times \Lambda \rightarrow \Theta$



LEARNER DEFINITION

In pseudo-code:

- Learner gets a hypothesis space of parametrized functions $\mathcal{H}$
- User passes data set $\mathcal{D}_{\text{train}}$ and control settings λ
- Learner sets parameters such that model fits data best
- Optimal parameters $\hat{\theta}$ or function $\hat{f}$ is returned for later usage

